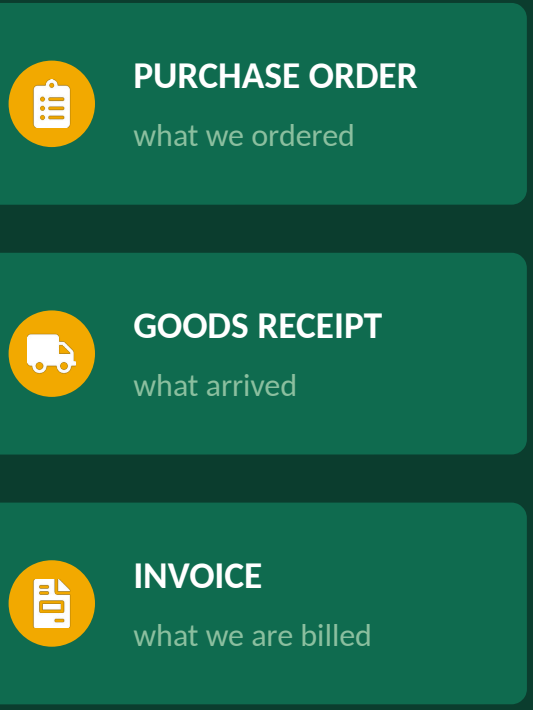
Agentic AI for Accounts Payable

Invoice three-way match with a team of cooperating agents

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Why invoices go wrong

Before paying a vendor, finance teams must check the invoice against the purchase order and the goods receipt. Done by hand, it is slow, and every miss becomes an overpayment.



Domain: Banking & Finance, Business Operations

What slips through

- **Price above the PO**
- **Billed more than received**
- **Same invoice paid twice**

From manual checks to agents

Today: manual review

- Staff compare three documents by hand
- Slow and repetitive for every invoice
- Errors often found after payment
- No consistent written reason

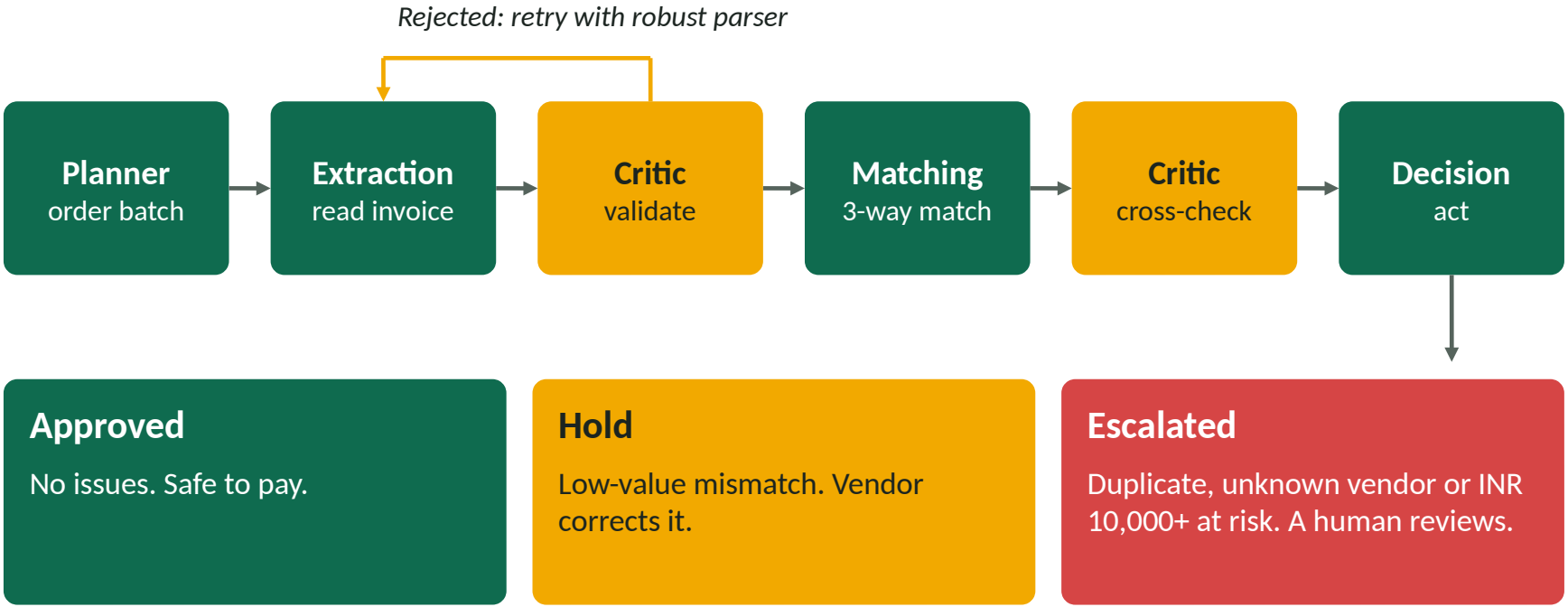
With agents

- Planner orders the batch by urgency
- Agents extract, validate and match
- Problems caught before payment
- Every decision comes with a reason

Five agents, one job

Planner	Sequences the batch by due date and amount; flags urgent invoices
Extraction	Turns invoice text into a typed schema with an LLM or parser; retries if rejected
Critic	Rejects extractions that fail arithmetic checks; confirms every variance is explained
Matching	Compares invoice, PO and goods receipt on price, quantity, tax, vendor and duplicates
Decision	Applies business rules to approve, hold or escalate, and writes the reason

How the agents hand off



System architecture

Streamlit UI
controls, live agent trace, results dashboard

LangGraph state graph
shared state, conditional retry edge, plain-Python fallback

Agents

Planner | Extraction | Critic | Matching | Decision

Tools and data
PO DB, GRN DB, vendor master, duplicate registry, rules

LLM
Ollama local model or mock

Data flow

1. Batch enters the UI
2. Graph passes shared state
3. Agents call tools and data
4. Decision returns to the UI

Demo walkthrough

1 Generate batch

24 synthetic invoices, 8 with seeded errors

2 Run agents

Watch each agent step live

3 Review decisions

Approved, held or escalated, colour-coded

4 Inspect

Raw invoice, trace, reason and metrics

Live trace for one invoice (real output)

```
Extraction Agent: attempt 1 via fast parser: INV-7015 | total 34.00
Critic Agent: extraction REJECTED (qty x unit != amount) -> retry
Extraction Agent: attempt 2 via robust parser: INV-7015 | total 34,673.17
Matching Agent: 3-way match done -> 1 issue(s): PRICE_MISMATCH
Decision Agent: HOLD (low-value mismatch: return to vendor for correction)
```

What the validation shows

8/8

seeded errors caught

8/8

issue types identified correctly

0/16

clean invoices wrongly flagged

1.77L

INR at risk flagged, equal to the seeded total

- Measured on 24 synthetic invoices: 8 seeded errors and 16 clean.
- The Critic loop caught misread amounts; the retry fixed them before matching.
- High scores reflect rule-based matching on known data, not real-world accuracy.

Limits and next steps

Limitations

- Synthetic data and one fixed invoice layout
- Rule-based matching scores optimistically
- No OCR, ERP link or user login
- Time savings are estimates, not measurements

Future scope

- OCR for scanned PDFs and photos
- ERP connectors such as SAP and Tally
- Benchmark LLM extraction against the parser
- Vendor fraud pattern detection
- Human approval screen and MCP tools

Thank you

Questions?



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